



The Price of Certainty: Verification Tiers and Their Cost

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Abstract

Verification is usually treated as a protocol constant: a system decides how work is checked and every participant lives with that choice. We argue it should be a purchased good with a published price curve, selected per job by the party bearing the loss.

We enumerate six verification tiers, give each an honest cost model, and derive the indifference conditions that separate them. The central quantity is the ratio of value at risk to verification cost, and it varies over many orders of magnitude across the jobs a real market will carry — which is exactly why a single protocol-wide soundness bar is either wasteful or unsafe, and usually both at once for different jobs.

We also make a refinement to the claim that pricing and verification are orthogonal. They are orthogonal in *mechanism* — the allocator need not know whether the provider is honest, and the verifier need not know whether the price was fair. They are coupled in the *objective*, because the cost of soundness is a term in the price. Getting this distinction right is what allows both to be designed independently without either being wrong.

Contents

1	Soundness as a demand-side parameter	3
2	The tiers	3
2.1	T0 — Recomputation by the buyer	3
2.2	T1 — Byte equality and known-answer tests	3
2.3	T2 — Freivalds and algebraic spot checks	3
2.4	T3 — Replication	4
2.5	T4 — Succinct proofs	4
2.6	T5 — Hardware attestation	4
3	Choosing a tier	4
4	Orthogonality, stated precisely	5
5	What a market should publish	5

1 Soundness as a demand-side parameter

A buyer posting a job faces an expected loss

$$\mathbb{E}[\text{loss}] = \pi \cdot (1 - \sigma) \cdot V, \quad (1)$$

where V is the value at risk if the answer is wrong, π is the prior probability that a given provider defects, and σ is the soundness of the verification tier — the probability that a defection is caught. Buying tier T costs $c(T)$. The buyer’s problem is

$$\min_T \left[c(T) + \pi (1 - \sigma(T)) V \right], \quad (2)$$

and the optimum plainly depends on V , which is private to the buyer and differs by six or more orders of magnitude between a chat completion and a settlement instruction. No protocol constant can be right for both.

Remark 1. *This is why we resist framing a market’s verification level as a property of the market. The market’s job is to make each tier available, priced, and honestly described. The choice belongs to whoever loses money when the answer is wrong.*

2 The tiers

2.1 T0 — Recomputation by the buyer

The buyer runs the job itself. Soundness is total; cost is the job. Rational only when the buyer wanted the answer for reasons other than lacking the capacity to compute it — audit sampling, spot checks at rate q , giving $\sigma = q$ at cost $q \cdot c_{\text{job}}$. Random audit at low q combined with a large stake is often the cheapest adequate mechanism and is systematically underused.

2.2 T1 — Byte equality and known-answer tests

Available only for determinism class D. The provider’s output is compared against a canonical implementation or a published vector set. Soundness is 1 against output corruption; cost is $O(1)$ amortised when vectors are published, or one canonical execution otherwise. Adjudication yields a proof of misbehaviour rather than a vote.

This tier is free in the only sense that matters — it costs nothing beyond what the market already publishes — and it is unavailable to anyone who has not done the work of making their kernels reproducible. That work is the subject of the companion paper.

2.3 T2 — Freivalds and algebraic spot checks

For the matrix products that dominate inference, correctness of $AB = C$ is checkable without recomputing the product. Draw r uniformly from $\{0, 1\}^n$ and test

$$A(Br) \stackrel{?}{=} Cr. \quad (3)$$

If $AB \neq C$ the test fails with probability at least $1/2$; after k independent rounds the soundness error is at most 2^{-k} .

The cost is three matrix–vector products, $6n^2$ operations, against $2n^3$ for the product itself. At $n = 4096$:

$$\frac{2n^3}{6n^2} = \frac{n}{3} = 1,365, \quad (4)$$

so a single round costs 0.073% of the product and twenty rounds — soundness error below 10^{-6} — costs 1.5%. For linear-algebra-heavy work this is the best value on the ladder by a wide margin, and it applies to class S where T1 does not.

Its limits should be stated. Freivalds checks the linear algebra, not the nonlinearities, the sampling, or that the weights were the ones paid for. It is a strong check on the expensive part of the computation and silent about the rest. Combining it with a weight commitment — the provider commits to a hash of the weight tensors, the buyer’s job specification names the expected hash — closes the most valuable of those gaps cheaply.

2.4 T3 — Replication

Dispatch to m independent providers, compare. With independent defection probability π and no collusion, undetected corruption requires all m to defect identically:

$$1 - \sigma = \pi^m \quad (\text{identical-defection model}), \quad c(T3) = m \cdot c_{\text{job}}. \quad (5)$$

Cost is linear in m ; soundness is exponential. The independence assumption is the weak point and it is not a technicality: providers running the same image on the same cloud in the same region are not independent, and a market that allocates purely on price will systematically produce correlated committees. Diversity has to be an explicit allocation constraint or replication silently degrades to $m = 1$.

2.5 T4 — Succinct proofs

The provider produces a proof π that the buyer verifies in time polylogarithmic in the computation. Soundness is cryptographic and independent of any honesty assumption. Prover overhead relative to native execution is currently 10^3 to 10^6 , depending on arithmetization and how badly the computation fits the field.

Rational when the same statement is verified many times, when verification must happen on-chain, or when V is large enough that $10^4 \times \text{compute}$ is still cheap insurance. For a single off-chain inference it is not close.

One property matters for anything intended to outlive the current cryptographic era: a proof system resting on pairings or discrete logarithms is Shor-breakable, which makes it an odd foundation for a market that markets itself on quantum readiness. Groth16, KZG and PLONK all fall in that class. Our proof substrate is deliberately restricted to hash-based STARK/FRI with no elliptic-curve recursion and no pairings anywhere in the pipeline, using cSHAKE256, KMAC256 and TupleHash256 (FIPS 202 with SP 800-185) as the hash family throughout so that Merkle commitments, FRI folding and the Fiat–Shamir transform present one auditable surface. We note plainly that this substrate is at a stable-surface, audit-gated stage rather than a finished one; it should be pinned to a tagged release and treated as forward-looking rather than deployed.

2.6 T5 — Hardware attestation

A trusted execution environment attests that the declared binary ran on unmodified hardware. Cost is near zero and soundness is high — against a different adversary. The trust root moves to a silicon vendor, and the history of side-channel and fault attacks on such environments argues against using them as a sole tier for high V . As a cheap complement to T2, raising the cost of the easiest attacks, they are good value.

3 Choosing a tier

Substituting into (2), tier T dominates tier T' when

$$c(T) - c(T') < \pi V [\sigma(T) - \sigma(T')]. \quad (6)$$

Tier	Cost / c_{job}	Soundness	Rational when
T1	≈ 0	1 (class D)	always, where available
T5	≈ 0	vendor-dependent	complement, never sole
T2	$0.001k$	$1 - 2^{-k}$	$\rho \gtrsim 10^2$, linear algebra
T0	q	q	$\rho \gtrsim 1/q$, with stake
T3	m	$1 - \pi^m$	$\rho \gtrsim 10^3$, diverse committee
T4	$10^3\text{--}10^6$	cryptographic	$\rho \gtrsim 10^6$, or on-chain

Table 1: Tier selection by value-to-compute ratio ρ .

Writing $\rho = V/c_{\text{job}}$ for the value-to-compute ratio, the boundaries take a simple form.

The practical reading: most volume should sit at T1 or T2, replication should be reserved for jobs whose value genuinely warrants tripling the bill, and proofs should be reserved for statements that are settled on-chain or reused. A market that defaults everything to replication is burning two thirds of its compute for soundness that most of its jobs do not need.

4 Orthogonality, stated precisely

The claim that pricing and verification are orthogonal is right and worth defending, but it needs a sharper statement than it usually gets.

Proposition 1 (Mechanism orthogonality). *The allocator requires no belief about provider honesty, and the verifier requires no belief about whether the clearing price was fair. Neither mechanism takes the other’s state as input.*

Proposition 2 (Objective coupling). *The cost of verification $c(T)$ appears as a term in the allocation objective. A buyer requiring T3 with a diversity constraint faces a different feasible set and a different clearing price than one requiring T1.*

Both are true simultaneously, and the distinction is what makes the architecture work: because the coupling is only through a scalar cost term, the two mechanisms can be designed, audited, replaced and reasoned about separately. That is a stronger and more useful statement than unqualified independence, which would be false — if verification were genuinely free of price consequences, no one would ever choose a cheaper tier.

5 What a market should publish

Three artifacts turn this from an analysis into infrastructure. A tier registry giving $c(T)$ and $\sigma(T)$ with stated assumptions, so buyers can compute their own optimum rather than trusting a recommendation. A per-provider capability declaration listing which tiers and determinism classes it honours, which the allocator treats as a hard constraint. And a public record of caught defections, which is what turns π from a guess into a measurement — without it, every buyer is solving (2) with an invented parameter.